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Real-Time FPGA-Based Pulse Simulator for the ATLAS Tile Calorimeter Readout Chain at the HL-LHC

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The LHC and the ATLAS experiment will undergo a Phase-II upgrade, imposing stringent requirements on real-time data processing. The Tile Calorimeter (TileCal) readout electronics will be fully replaced, and the digitised samples of every channel will be streamed continuously at the LHC bunch-crossing frequency of 40.08 MHz to off-detector FPGAs, where the deposited energy is reconstructed online. Developing, validating and testing the firmware that performs this reconstruction calls for a real-time FPGA-based simulator of the TileCal readout chain, which has been built for this purpose. It generates, at every bunch crossing, the energy deposited in a selected calorimeter cell according to its statistical distribution, adds realistic electronic noise, reproduces the response of the FENICS front-end shaper by means of an IIR filter, and emulates the 12-bit ADC of the upgraded electronics. Implemented on a Kintex UltraScale device, it uses about 1 % of the available logic and closes timing with about 3 % of margin over the bunch-crossing frequency. This enables systematic validation, stress testing and performance evaluation of hardware-based trigger and energy reconstruction firmware under realistic and repeatable conditions.

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1. Introduction

Simulation is a pillar of experimental high-energy physics, used to design detectors, validate reconstruction algorithms and interpret the recorded data. In ATLAS [1] this relies on a Monte Carlo chain based on GEANT4 [2] and on digitisation models of the readout electronics [3], running offline on CPU-based computing farms, increasingly with GPU acceleration. Such simulations are accurate but cannot deliver a continuous stream of detector signals synchronously with the collision rate, as is needed to exercise firmware that processes every bunch crossing as it arrives.

This matters for the High-Luminosity LHC (HL-LHC) [4], where the ATLAS Tile Calorimeter (TileCal) [5] will operate with completely new readout electronics [6]. The upgraded front-end no longer stores the digitised samples on detector awaiting a trigger decision: the samples of all channels are streamed off-detector at the LHC bunch-crossing rate of 40.08 MHz and their energy is reconstructed in real time, at every bunch crossing, in the FPGAs of the Tile PreProcessor (TilePPr) [7]. Energy reconstruction thus becomes a permanently running streaming system whose performance and stability directly affect the trigger data, motivating the development of online reconstruction methods, from optimal filtering [8] to neural networks on FPGAs [9].

Evaluating such methods requires stimuli delivered at the rate the detector produces them: the estimator is itself firmware, which has to meet timing at the bunch-crossing rate and whose output depends on the samples just processed, so slower pre-computed data test the arithmetic but not the implementation. A real-time pulse simulator of the TileCal readout chain, implemented in an FPGA and generating data at that rate, can feed the firmware under test so that the performance and the stability of a new estimator can be assessed while the pulses pile up and the ADC saturates, over runs far longer than any stored dataset would allow. Hardware pulse generators in nuclear instrumentation pose similar challenges [10]. Both the deposited energy and the electronic noise are stochastic, so the firmware must embed uniform Pseudo-Random Number Generators (pRNGs) that are statistically uncorrelated [11] and a mechanism shaping them into arbitrary target distributions with modest hardware usage [12]. The front-end response and the ADC must also be simulated. This contribution describes the first version of the Real-Time Pulse Simulator, applied to one of the most exposed TileCal cells.

2. The readout chain at the HL-LHC and the simulator architecture

TileCal is the central hadronic calorimeter of ATLAS [5]: steel absorbers and scintillating tiles read out by wavelength-shifting fibres coupled to Photomultiplier Tubes (PMTs). Figure 1 shows the readout chain foreseen for the HL-LHC [6]. The FENICS front-end board splits the PMT signal into a High-Gain (HG) and a Low-Gain (LG) branch and shapes it into a pulse spanning several bunch crossings; both branches are digitised by 12-bit ADCs located on the mainboard. The daughterboard serialises the data from up to 12 photomultipliers and transmits them over 9.6 Gbit s^{-1} links to the TilePPr, where the Compact Processing Modules (CPMs) reconstruct the pulse amplitude (proportional to the deposited energy) and build the digital trigger sums.

The simulator reproduces the on-detector part of this data path in firmware, from the energy deposited in the cell to the samples arriving at the TilePPr input; the TilePPr and the amplitude reconstruction of the CPMs are what it is built to exercise, and are deliberately left outside it. Each

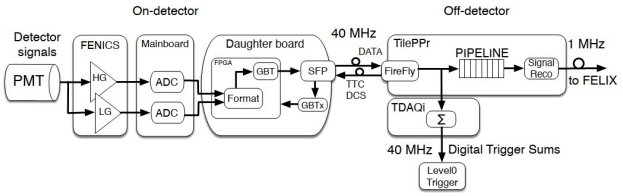


Figure 1: Readout chain of the ATLAS Tile Calorimeter for the HL-LHC [6].

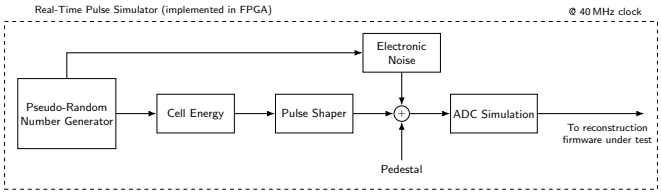


Figure 2: Block diagram of the Real-Time Pulse Simulator, reproducing in firmware the data path from the cell energy to the digital samples entering the TilePPr, synchronously with the bunch-crossing clock. Original figure, produced for this contribution.

block maps onto an element of the chain (figure 2): a uniform pRNG drives the *Cell Energy* block, which converts uniform numbers into energies distributed according to the occupancy of the simulated cell. The *Pulse Shaper* block reproduces the FENICS shaping stage; an independent instance of the generator produces the *Electronic Noise*, added to the shaped signal with a programmable *Pedestal*; and the *ADC Simulation* block emulates the digitisation, delivering one sample per bunch crossing to the CPM firmware under test. The reference model is the Tile Pulse Simulator, an already existing ATLAS package: the statistical description of the cell energy is taken from it and the firmware output validated against it, whereas the front-end pulse shape comes from a SPICE simulation of the FENICS schematic (section 4). A single channel is simulated, the A13 high-gain channel, an A-layer cell of the extended barrel whose innermost radial position gives one of the highest occupancies in TileCal.

3. Cell energy and electronic noise generation

The energy deposited in a cell at each bunch crossing is a random variable whose distribution depends on the cell position and on the pile-up conditions; generating it in hardware requires uniform random numbers and a transformation onto the target distribution. The uniform source is a pRNG based on Linear Feedback Shift Registers (LFSRs). A free-running LFSR produces strongly correlated neighbouring samples, unacceptable when the numbers emulate independent events; the architecture adopted here therefore uses seven parallel 42-bit LFSRs with different seeds, selected by a multiplexer, giving an essentially impulsive autocorrelation at low cost [11].

The mapping onto the target distribution uses the Inverse Cumulative Distribution Function (ICDF) method: a uniform number addresses a Look-Up Table (LUT) holding the discretised ICDF, so that its output follows the target statistics. A single LUT samples uniformly in probability and resolves poorly the high-energy tail that matters for saturation and trigger studies. The multi-memory approach of Ref. [12] is used instead: the CDF is segmented into regions, each covered by its own low-depth memory addressed by an independent uniform generator (figure 3, left). Resolution thus grows only where needed, with a small memory footprint and no DSP blocks.

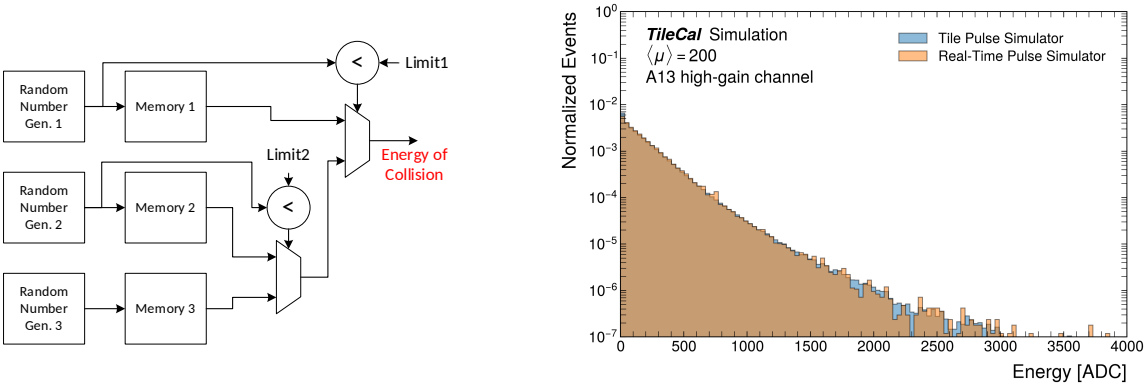


Figure 3: Left: multi-memory implementation of the ICDF, taken from Ref. [12]; each memory covers one region of the CDF and is addressed by an independent uniform generator, and the comparators select which memory drives the output. Right: comparison of the truth-energy distributions generated for the TileCal A13 high-gain channel by the Tile Pulse Simulator and by the FPGA-based Real-Time Pulse Simulator, before the detector readout chain. Plot approved by the ATLAS Collaboration, published here for the first time.

Here the target is the truth energy of the A13 TileCal cell recorded in minimum-bias events simulated under HL-LHC pile-up conditions at $\langle\mu\rangle = 200$. Figure 3 (right) compares it with the Tile Pulse Simulator reference: the agreement extends over more than six orders of magnitude in probability and over the full energy range. The same technique generates the electronic noise, with the target distribution given by a Gaussian model of the channel noise.

4. FENICS shaper simulation and ADC emulation

The FENICS front-end board amplifies and shapes the PMT current pulse before digitisation. Its long tail generates the pile-up and baseline shift the online algorithms must handle. A Finite Impulse Response (FIR) filter would need a prohibitively high order for such a slowly decaying response, so an Infinite Impulse Response (IIR) implementation was adopted: the nominal analogue response, from a SPICE simulation of the front-end schematic, was discretised at the 25 ns sampling period and split into six parallel first- and second-order sections in fixed-point arithmetic.

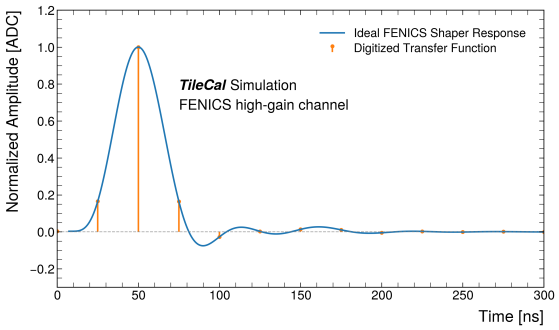


Figure 4: Normalised nominal impulse response of the FENICS high-gain shaper, obtained from a SPICE-based circuit simulation, compared with the digitised transfer function implemented in fixed-point arithmetic in the simulator through an IIR filter architecture, assuming ideal sampling timing. Plot approved by the ATLAS Collaboration, published here for the first time.

Figure 4 compares the fixed-point firmware output, obtained from a bit-accurate simulation of the IIR filter as implemented, with the nominal analogue pulse shape of the high-gain channel,

for sampling instants aligned with the pulse peak, i.e. without phase deviation between collision and sampling clock. Both waveforms are normalised to unit peak amplitude and compared sample by sample: the largest deviation, relative to that peak, is 1.25×10^{-5} %, below 10^{-3} of one least-significant bit for a pulse spanning the full ADC range. The shaped signal is not yet identical to the signal received by the TilePPr, because the real ADC has a finite offset (pedestal) and dynamic range: the ADC emulation block adds the noise and a configurable pedestal and clips the result to the interval $[0, 4095]$ ADC counts of the 12-bit ADCs, reproducing saturation and baseline undershoot as the real reconstruction would see them.

5. FPGA implementation and results

The simulator was implemented on an AMD Kintex UltraScale KCU105 kit, whose device belongs to the same family as the FPGAs of the CPMs in the TilePPr. The design uses 2603 look-up tables (1.07 % of the device), 2400 registers (0.50 %), 7 DSP blocks (0.36 %) and no block RAM, since the ICDF tables fit in distributed memory. It closes timing at 41.2 MHz, about 3 % above the bunch-crossing frequency. Extrapolating from the logic utilisation alone, of the order of 90 channels would fit in one device; routing and timing closure, not logic, should set the practical limit of a multi-channel design.

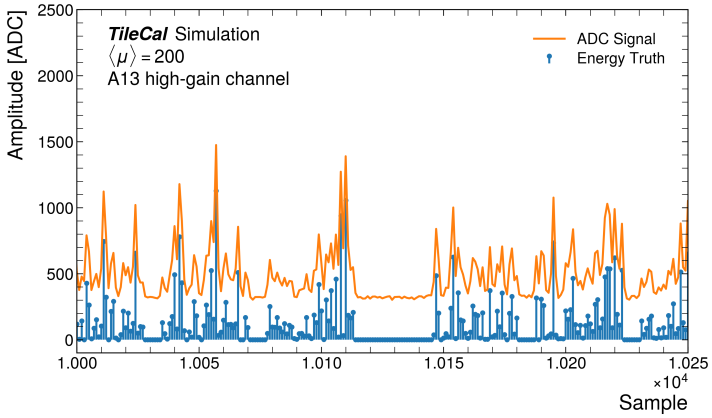


Figure 5: Truth-energy sequence generated for the TileCal A13 cell together with the corresponding ADC output produced by the FPGA-based Real-Time Pulse Simulator after the complete detector readout chain. The horizontal axis is the bunch-crossing number, one unit being 25 ns. The ADC waveform is offset from the truth energy by the pedestal, added to every sample. Periods without collisions appear as beam gaps and follow the filling scheme of the simulated run. Plot approved by the ATLAS Collaboration, published here for the first time.

Figure 5 shows the data stream produced for the A13 high-gain channel at $\langle\mu\rangle = 200$: blue markers are the truth energies, the orange curve the corresponding ADC output after shaping, noise and pedestal addition and saturation. The ADC waveform never returns to the truth-energy baseline but sits on the programmed pedestal, 500 ADC counts here, and rises above it whenever the local collision activity increases. Neither effect is an artefact: the pedestal is added on purpose, so that the undershoot of the shaped pulse stays inside the ADC range, and the slow rise is the pile-up baseline shift, from the superposition of the long pulse tails, that the algorithms have to cope with. Isolated deposits produce a clear peak, whereas consecutive ones merge and their amplitudes are no longer separable. The flat region with zero truth energy is a gap between bunch trains.

Since both quantities are available in real time, the bias and resolution of a new estimator can be measured sample by sample against the known input, versus the local pile-up.

6. Summary

A real-time pulse simulator of the TileCal readout chain for the HL-LHC has been successfully implemented and operated on the KCU105 board at the target bunch-crossing rate. It reproduces the whole on-detector chain, from the cell energy to the 12-bit ADC, using about 1 % of the device resources. The generated distributions agree with the reference Tile Pulse Simulator, making it a realistic and repeatable platform for evaluating online reconstruction algorithms; it has already been used in studies of online energy reconstruction within the TileCal group.

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